

QUIZMASTER-V1

A MODERN APPLICATION DEVELOPMENT-1 PROJECT

JAN 2025 TERM

PROJECT BY:

AMAN KUMAR RAWAT

23F2001035

STUDENT DETAILS:

NAME: AMAN KUMAR RAWAT

ROLL NO: 23F2001035

EMAIL: 23F2001035@ds.study.iitm.ac.in

ABOUT ME :

I am a **19 years old student** with a strong passion for both **mechanical engineering and technology**. Currently, I am pursuing **BS degree in Data Science from IIT Madras currently in the Diploma Level**, along with that I am also pursuing **Btech in MECHANICAL ENGINEERING**. My curiosity lies at the intersection of **engineering principles and computational techniques**, where I enjoy exploring how data-driven solutions can enhance mechanical systems and real-world applications. I am always eager to **learn, innovate, and apply my knowledge** to challenging technical projects.

Technologies Used:

- Flask framework - for development of the application.
- Jinja2- for dynamic templating and HTML generation.
- Bootstrap- for styling and designing purpose.
- Flask-SQLAlchemy and sqlite3- for database operations.
- Matplotlib - for generating the the admin_summary and user_summary.

How I approached the problem:

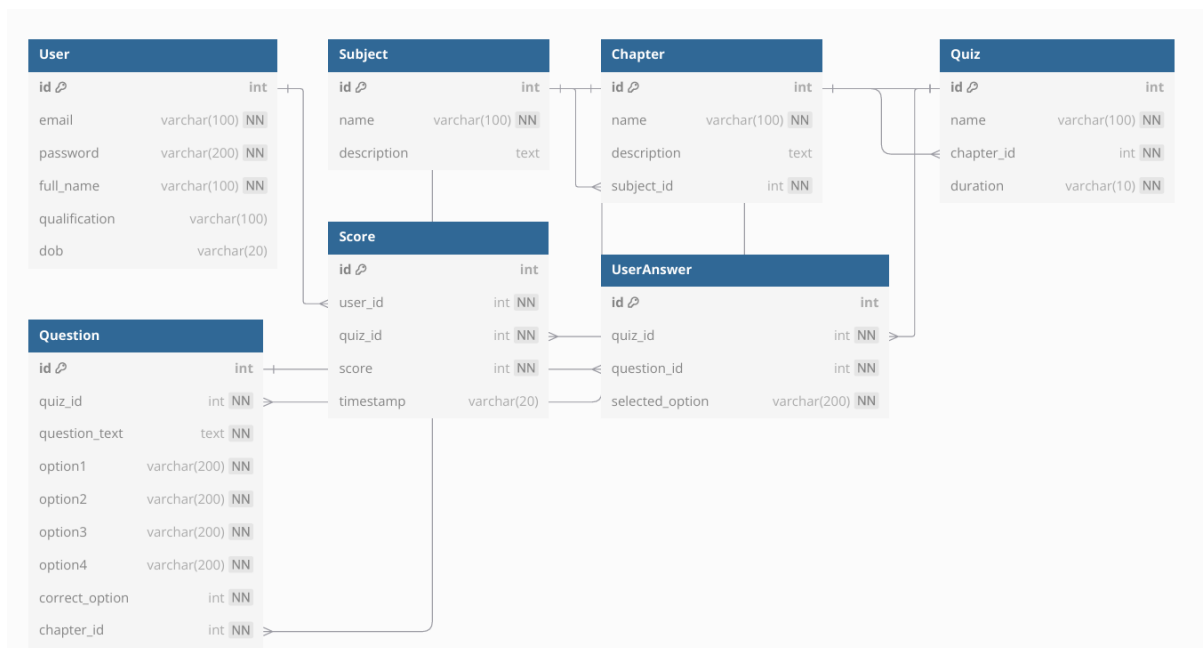
Without having any prior experience in development, I was so nervous that whether I will be

Able to complete the project or not. I first started with creating templates. Which helped me understand

The basic structure of the application. I took inspiration from the wireframe provided in the guideline document. After that, I watched the mad1 project playlist that are available on the mad1 youtube channel of IITM BS DEGREE. Which played a crucial role for me to understand the concept of flask and the CRUD operations. Now I was good to go and I started working on the project with full dedication. I also watched

Several youtube channel to learn flask like codewithharry. This project helped me gaining confidence, knowledge and moreover power of consistency and hardwork.

DB SCHEMA:



1. User Table

- Stores user information such as email, password, full name, and qualification details.
- **Primary Key:** id
- **One-to-Many Relationship:** A user can have multiple quiz scores.

2. Subject Table

- Represents different subjects under which quizzes are categorized.
- **Primary Key:** id
- **One-to-Many Relationship:** A subject can have multiple chapters.

3. Chapter Table

- Represents chapters under each subject, helping in organizing quizzes topic-wise.
- **Primary Key:** id
- **Foreign Key:** subject_id (References Subject Table)
- **One-to-Many Relationship:** A chapter can have multiple quizzes.

4. Quiz Table

- Stores quiz-related information, including name and duration.
- **Primary Key:** id
- **Foreign Key:** chapter_id (References Chapter Table)
- **One-to-Many Relationship:** A quiz contains multiple questions.

5. Question Table

- Stores quiz questions along with multiple-choice options and the correct answer.
- **Primary Key:** id
- **Foreign Keys:** quiz_id, chapter_id (References Quiz & Chapter Tables)
- **One-to-Many Relationship:** A quiz can have multiple questions.

6. Score Table

- Records users' quiz scores and timestamps.
- **Primary Key:** id
- **Foreign Keys:** user_id, quiz_id (References User & Quiz Tables)
- **One-to-Many Relationship:** A user can have multiple score entries.

7. UserAnswer Table

- Stores users' responses to quiz questions.
- **Primary Key:** id
- **Foreign Keys:** quiz_id, question_id (References Quiz & Question Tables)
- **One-to-Many Relationship:** A user can submit multiple answers per quiz attempt.

VIDEO LINK: [click here to view](#)